

SPECIFICATION OF
D/S : Flugal DS DU (Malaysia)
Dimension : L 284 mm x W 148 mm
Paper GSM : 60 GSM Indonesian Offset
Printing : Single Color (Both side print)

Flugal
Fluconazole EP

1. Name of the Medicinal Product: *Flugal*
2. Qualitative and Quantitative composition
Flugal 150 capsule : Each capsule contains Fluconazole EP 150 mg.

Fluconazole is a bis-triazole, having the following chemical name: 2-(2,4-difluorophenyl)-1,3-bis(1H,2,4-triazol-1-yl)-2 propanol. It has a molecular weight of 306.3.

3. Pharmaceutical Form

Capsule

Product description: A hard gelatin capsule (Size#2) filled with white powder having a blue color cap and creamish color body imprinted "Square" on both parts.

Route of administration : Oral

4. Clinical particulars

4.1 Therapeutic indications

Therapy may be instituted before the results of the cultures and other laboratory studies are known; however, once these results become available, anti-infective therapy should be adjusted accordingly.

1. Cryptococcosis, including cryptococcal meningitis and infections of other sites (e.g., pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplants or other causes of immunosuppression may be treated. *Flugal* can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.
2. Systemic candidiasis, including candidemia, disseminated candidiasis and other forms of invasive candidal infection. These include infections of the peritoneum, endocardium, eye and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated.
3. Mucosal candidiasis, these include oropharyngeal, esophageal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated. Prevention of relapse of oropharyngeal candidiasis in patients with AIDS.
4. Genital candidiasis, vaginal candidiasis, acute or recurrent, and prophylaxis to reduce the incidence of recurrent vaginal candidiasis (3 or more episodes a year), candidal balanitis.
5. Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.
6. Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, tinea versicolor, and dermal candida infections.

4.2 Posology and Method of Administration

The daily dose of *Flugal* should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single dose therapy. Therapy for those types of infections requiring multiple dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An inadequate period of treatment may lead to recurrence of active infection. Patients with AIDS and cryptococcal meningitis or recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.

Use in Adults:

1. For cryptococcal meningitis and cryptococcal infections at other sites, the usual dose of *Flugal* is 400 mg on the first day followed by 200 to 400 mg once daily. Duration of treatment for cryptococcal infection will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis.
2. For the prevention of relapse of cryptococcal meningitis in patients with AIDS, after the patient receives a full course of primary therapy, *Flugal* may be administered indefinitely at a daily dose of 200 mg.
3. For candidemia, disseminated candidiasis and other invasive candidal infections, the usual daily dose of *Flugal* is 400 mg on the first day followed by 200 mg daily. Depending on the clinical response, the dose may be increased to 400 mg daily. Duration of treatment is based upon the clinical response.
4. For oropharyngeal candidiasis, the usual dose of *Flugal* is 50 to 100 mg once daily for 7-14 days. If necessary, treatment can be continued for longer periods in patients with severely compromised immune function. For atrophic oral candidiasis associated with dentures, the usual dose is 50 mg once daily for 14 days administered concurrently with local antiseptic measures to the denture.

For other candidal infections of mucosa except genital candidiasis (e.g. esophagitis, non-invasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis, etc.) the usual effective dose is 50 to 100 mg daily, given for 14-30 days.

For the prevention of relapse of oropharyngeal candidiasis in patients with AIDS, after the patient receives a full course of primary therapy, *Flugal* may be administered at a 150 mg once weekly dose.

4. For the treatment of vaginal candidiasis, *Flugal* 150 mg should be administered as a single oral dose.

To reduce the incidence of recurrent vaginal candidiasis, a 150 mg once monthly dose may be used. The duration of therapy should be individualized, but ranges from 4-12 months. Some patients may require more frequent dosing. For Candida balanitis, fluconazole 150 mg should be administered as a single oral dose.

5. The recommended *Flugal* dosage for the prevention of candidiasis is 50 to 400 mg once daily, based on the patient's risk for developing fungal infection. For patients at high risk of systemic infection, e.g. Patients who are anticipated to have profound or prolonged neutropenia, the recommended daily dose is 400 mg once daily. *Flugal* administration should start several days before the anticipated onset of neutropenia and continue for 7 days after the neutrophil count rises above 1000 cells per mm³.
 6. For dermal infections including tinea pedis, corporis, cruris and candida infections the recommended dosage of *Flugal* is 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2 to 4 weeks but tinea pedis may require treatment for up to 6 weeks.
- For tinea versicolor, the recommended dose is 300 mg once weekly for 2 weeks; a third weekly dose of 300 mg may be needed in some patients, whereas, in some patients, a single dose of 300 to 400 mg may be sufficient. An alternate dosing regimen is 50 mg once daily for 2 to 4 weeks.

Use in Elderly:

Where there is no evidence of renal impairment, normal dosage recommendation should be adopted. For patients with renal impairment (creatinine clearance <50 ml/min) the dosage schedule should be adjusted as described below.

Use in Renal Impairment:

Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of fluconazole an initial loading dose of 50 to 400 mg should be given. After the loading dose, the daily dose (according to indication) should be based on the following table:

mothers were treated with high-dose (400mg/day to 800mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5mg/kg or 10 mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 25mg/kg and 50mg/kg and higher doses. At doses ranging from 80mg/kg 320mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg, embryolethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification.

Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

Use During Lactation

Fluconazole is found in human breast milk at concentrations similar to plasma. Breast-feeding may be maintained after a single dose of 150mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole.

4.7 Effects on Ability to Drive and Use Machines

Experience with fluconazole indicates that therapy is unlikely to impair a patient's ability to drive or use machinery.

4.8 Undesirable effects

Fluconazole is generally well tolerated. The most common undesirable effects associated with fluconazole are:

Nervous System Disorders : Headache.

Skin and Subcutaneous Tissue Disorders: Rash.

Gastrointestinal Disorders: Abdominal pain, diarrhoea, flatulence, nausea. In some patients, particularly those with serious underlying diseases such as AIDS and cancer, changes in renal and haematological function test results and hepatic abnormalities have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain.

Hepatobiliary Disorders: Hepatic toxicity including rare cases of fatalities, elevated alkaline phosphatase, elevated bilirubin, elevated SGOT, elevated SGPT.

In addition, the following undesirable effects have occurred during post-marketing:

Nervous System Disorders: Dizziness, seizures, taste perversion.

Skin and Subcutaneous Tissue Disorders: Alopecia, exfoliative skin disorders including Stevens-Johnson syndrome and toxic epidermal necrolysis.

Gastrointestinal Disorders: Dyspepsia, vomiting.

Blood and Lymphatic System Disorders: Leukopenia including neutropenia and agranulocytosis, thrombocytopenia.

Immune System Disorders: Allergic reaction - Anaphylaxis (including angioedema, face oedema, pruritus), urticaria.

Hepatobiliary Disorders: Hepatic failure, hepatitis, hepatocellular necrosis, jaundice.

Metabolism and Nutrition Disorders: Hypercholesterolaemia,

hypertriglyceridaemia, hypokalaemia.

Cardiac Disorders: QT prolongation, torsade de pointes.

4.9 Overdose

There has been one reported case of overdosage with fluconazole. A 42-year-old patient infected with human immunodeficiency virus developed hallucinations and exhibited paranoid behavior after reportedly ingesting 8200 mg of fluconazole. The patient was admitted to the hospital, and his condition resolved within 48 hours.

Creatinine Clearance (ml/min)

>50 100% or 24 hours
<50 (no dialysis) 50% or 48 hours
Regular dialysis 100% after each dialysis

These are suggested dose adjustments based on pharmacokinetics following administration of single doses. Further adjustment may be needed depending on clinical condition.

Administration

Fluconazole may be administered either orally or by intravenous infusion at a rate not exceeding 10 ml/minute, the route being dependent on the clinical state of the patient. On transferring from the intravenous to the oral route, or vice versa there is no need to change the daily dosage.

4.3 Contraindications

Fluconazole should not be used in patients with known sensitivity to the drug, any of the inert ingredient or to related azole compounds.

Coadministration of terfenadine is contraindicated in patients receiving fluconazole at multiple doses of 400 mg per day or higher based upon results of a multiple dose interaction study (see section 4.5-Interaction with Other Medicaments and Other Forms of Interaction)

Coadministration of cisapride is contraindicated in patients receiving fluconazole (see Section 4.5 Interaction with Other Medicaments and Other Forms of Interaction).

Fluconazole is generally unadvised in association with halofantrine.

Fluconazole SHOULD NOT BE ADMINISTERED in the following cases:

- hypersensitivity to fluconazole and/or other azole derivatives
- pregnancy and lactation (See section 4.6 - Pregnancy and Lactation)
- In combination With cisapride and pimozide (see section 4.5 - Interaction with Other Medicaments and Other Forms of Interaction)

4.4 Special Warnings and Special Precautions for Use

Fluconazole has been associated with rare cases of serious hepatic toxicity including fatalities, primarily in patients with serious underlying medical conditions. In cases of fluconazole-associated hepatotoxicity, no obvious relationship to total daily dose, duration of therapy, sex or age of patient has been observed. Fluconazole hepatotoxicity has usually been reversible on discontinuation of therapy. Patients who develop abnormal liver function tests during fluconazole therapy should be monitored for the development of more serious hepatic injury. Fluconazole should be discontinued if clinical signs or symptoms consistent with liver disease develop that may be attributable to fluconazole.

Patients have rarely developed exfoliative cutaneous reactions, such as Stevens-Johnson Syndrome and toxic epidermal necrolysis, during treatment with fluconazole. AIDS patients are more prone to the development of severe cutaneous reactions to many drugs. If a rash, which is considered attributable to fluconazole, develops in a patient treated for a superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive/systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop.

The coadministration of fluconazole at doses lower than 400 mg per day with terfenadine should be carefully monitored.

In rare cases, as with other azoles, anaphylaxis has been reported.

Some azoles, including fluconazole, have been associated with prolongation of the QT interval on the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT prolongation and torsade de pointes in patients taking fluconazole. These reports included seriously ill patients with multiple confounding risk factors, such as structural heart disease, electrolyte abnormalities and concomitant medications that may have been contributory. Fluconazole should be administered with caution to patients with these potentially proarrhythmic conditions.

4.5 Interaction with Other Medicaments and Other forms of Interaction

Anticoagulants:

In an interaction study, fluconazole increased the prothrombin time (12%) after warfarin administration in healthy males. In post-marketing experience, as with other azole antifungals, bleeding events (bruising, epistaxis, gastrointestinal bleeding, hematuria and melena) have been reported. In association with increases in prothrombin time in patients receiving fluconazole concurrently with warfarin. Prothrombin time in patients receiving coumarin-type anticoagulants should be carefully monitored.

Azithromycin: An open-label, randomized, three-way crossover study in 18 healthy subjects assessed the effect of a single 1200 mg oral dose of azithromycin on the pharmacokinetics of a single 800 mg oral dose of fluconazole as well as the effects of fluconazole on the pharmacokinetics of Azithromycin. There was no significant pharmacokinetic interaction between fluconazole and azithromycin.

Benzodiazepines (Short acting): Following oral administration of midazolam, fluconazole resulted in substantial increases in midazolam concentrations and psychomotor effects. This effect on midazolam appears to be more pronounced following oral administration of fluconazole than with fluconazole administered intravenously. If concomitant benzodiazepine therapy is necessary in patients being treated with fluconazole, consideration should be given to decreasing the benzodiazepine dosage, and the patients should be appropriately monitored.

Cisapride: There have been reports of cardiac events including torsade de pointes in patients to whom fluconazole and cisapride were coadministered. Cisapride increased risk of ventricular arrhythmia troubles notably torsades de pointes. A controlled study found that concomitant fluconazole 200 mg once daily and cisapride 20 mg four times a day yielded a significant increase in cisapride plasma levels and prolongation of QTc interval. Coadministration of cisapride is contraindicated in patients receiving fluconazole.

Cyclosporin: A kinetic study in renal transplant patients found fluconazole 200 mg daily to slowly increase cyclosporin concentrations. However, in another multiple dose study with 100 mg daily, fluconazole did not affect cyclosporin levels in patients with bone marrow transplants. Cyclosporin plasma concentration monitoring in patients receiving fluconazole is recommended.

Hydrochlorothiazide : In a kinetic interaction study, coadministration of multiple-dose hydrochlorothiazide to healthy volunteers receiving fluconazole increased plasma concentrations of fluconazole by 40%. An effect of this magnitude should not necessitate a change in the fluconazole dose regimen in subjects receiving concomitant diuretics, although the prescriber should bear it in mind.

Oral contraceptives: Three kinetic studies with a combined oral contraceptive have been performed using multiple doses of fluconazole. There were no relevant effects on either hormone level in the 50 mg fluconazole study, while at 200 mg daily, the AUCs of ethinyl estradiol and levonorgestrel were increased 40% and 24% respectively. In a 300 mg once weekly fluconazole study, the AUCs of ethinyl estradiol and norethindrone were increased by 24% and 13% respectively. Thus, multiple dose use of fluconazole at those doses is unlikely to have an effect on the efficacy of the combined oral contraceptive.

Phenytoin: Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to a clinically significant degree. If it is necessary to administer both drugs concomitantly, phenytoin level should be monitored and the phenytoin dose adjusted to maintain therapeutic levels.

Pimozide: Combination with fluconazole will increase risk of ventricular arrhythmia troubles notably torsades de pointes.

Rifabutin: There have been reports that an interaction exists when fluconazole is administered concomitantly with rifabutin, leading to increased serum levels of rifabutin. There have been reports of uveitis in patients to whom fluconazole and rifabutin were coadministered. Patients receiving rifabutin and fluconazole concomitantly should be carefully monitored.

Rifampicin: Concomitant administration of fluconazole and rifampicin resulted in a 25% decrease in the AUC and a 20% shorter half-life of fluconazole.

circulating metabolites.

The long plasma elimination half-life provides the basis for single dose therapy for vaginal candidiasis, once daily and once weekly dosing for other indications.

6. Pharmaceutical Particulars

6.1 List of excipients

Capsule: Maize starch BP, Magnesium stearate BP, purified Talc BP, Empty Hard Gelatin Capsule (Ph. Grade)

6.2 Special precaution for Storage

Store at temperature not exceeding 30°C. Keep medicine out of reach of children. JAUHKAN DARI KANAK-KANAK.

6.3 Nature and Contents of Container

Capsules

Alu-Alu blister packing.

6.4 Instructions for Use/Handling

Capsules should be swallowed wholly.

6.5 Pack size

Flugal 150 capsule : Each small box contains 1x1 capsule in Alu-Alu blister pack & each large box contains 10 small boxes.

Manufactured by :



**SQUARE
PHARMACEUTICALS LTD.**
Kaliakoir, Gazipur, Bangladesh

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